Properties of the 2x2x2 Rubik's Cube in Graph Theory

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MATH 1030 Graph Theory

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Motivation

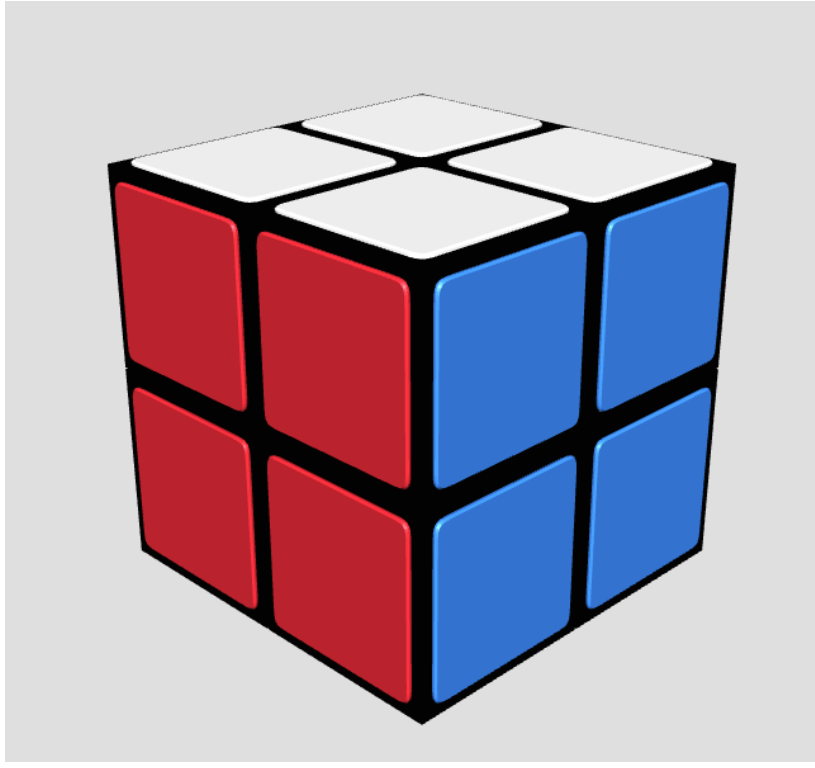
The Rubik's Cube has a niche and rich history on college campuses.

- 1980: MIT CubeLovers Email List. World's first online forum for discussing math and Rubik's Cube.
- 1982: Yale Math Department offers undergraduate seminar on "The Magic Cube."
- 2004: Caltech Cube Club. Organized first modern competitions.
- 2006: Berkeley Cube Club. "DeCal" is a Pass-Fail course teaching group theory and algorithms applied to the cube.
- 2010: Stanford researchers proved God's Number using 35 CPU-years of compute donated from Google.

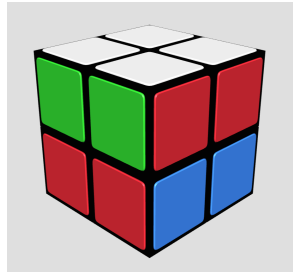
Roadmap

- **Foundation**
  - Group Notation and Cayley Graphs
- **Hamiltonian Properties**
  - The Hamiltonian Path Problem
  - Bipartite Structure and Hamiltonian Laceability
- **Vertex Transitivity**

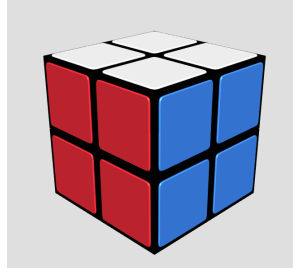
Cayley Graph: Local Neighborhood



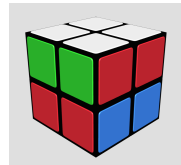
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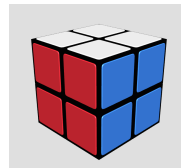
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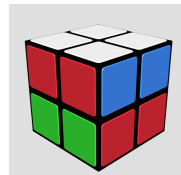
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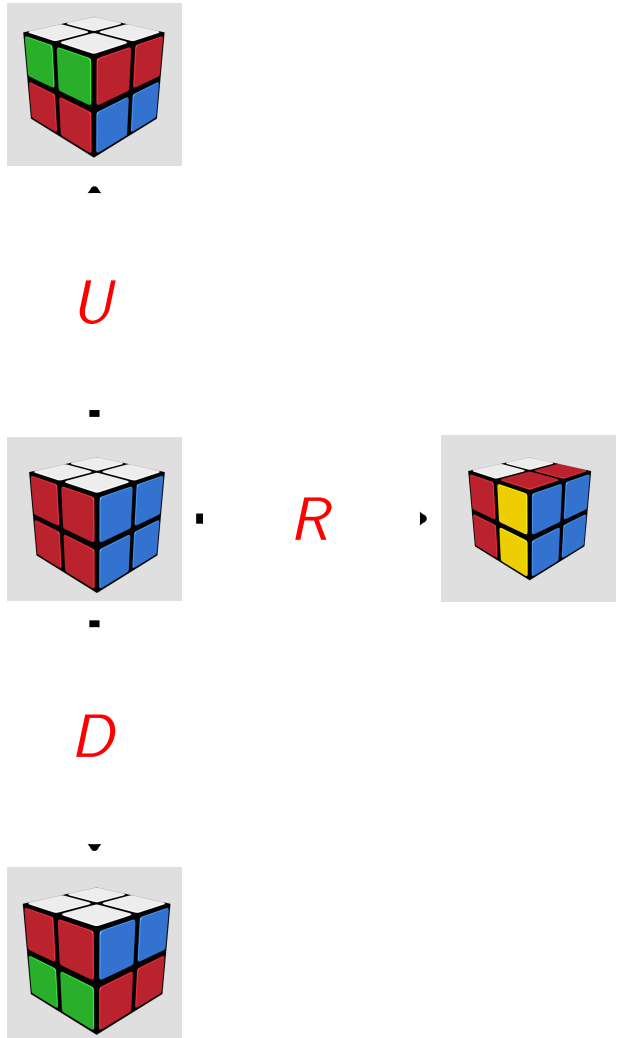
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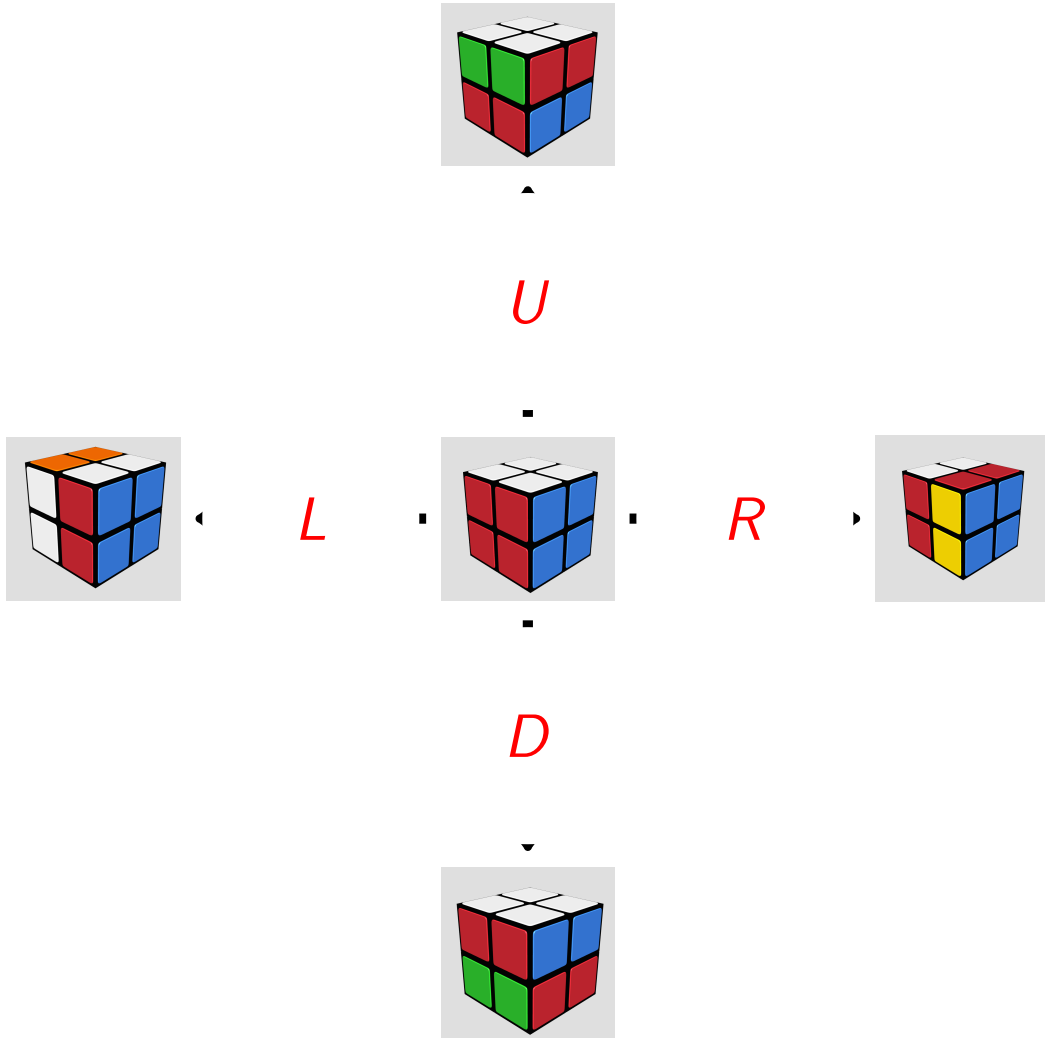
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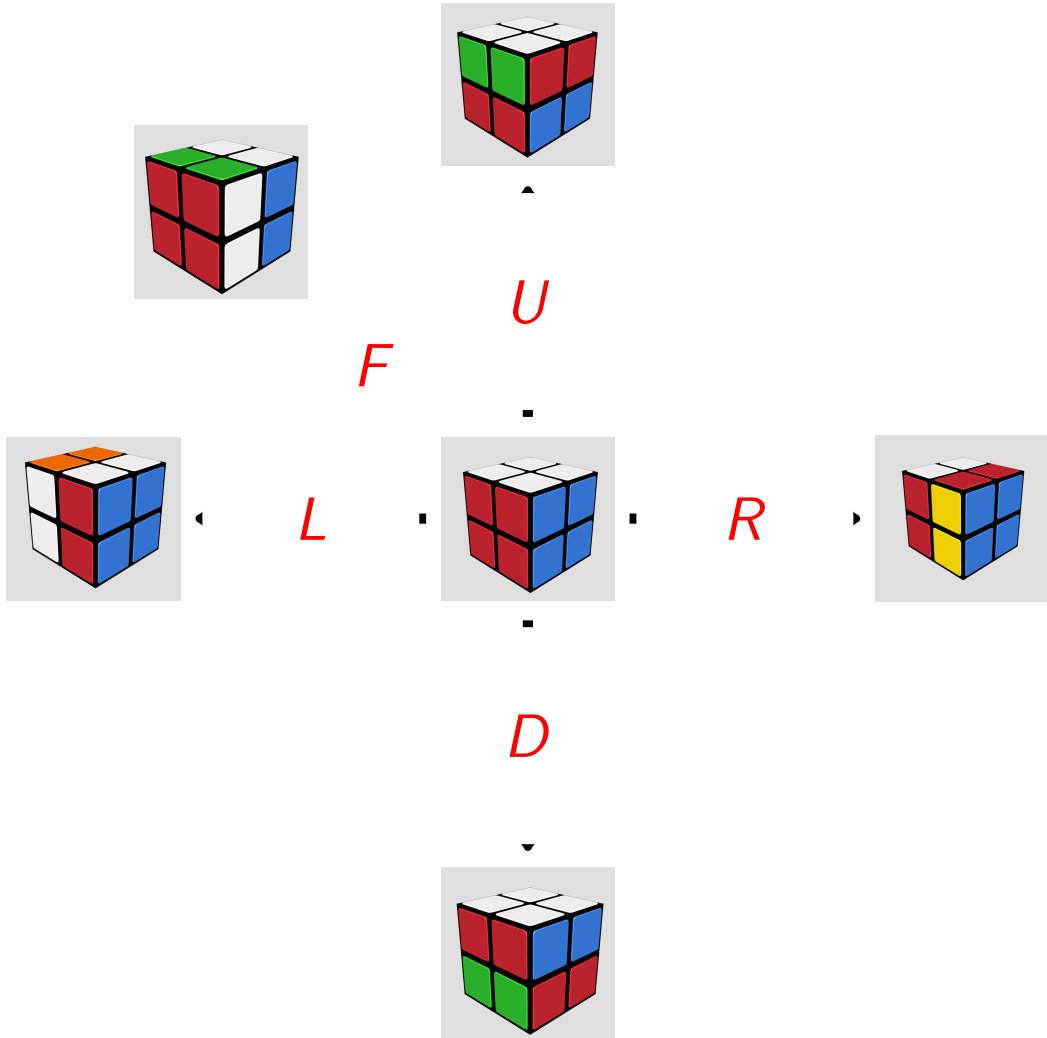
Cayley Graph: Local Neighborhood



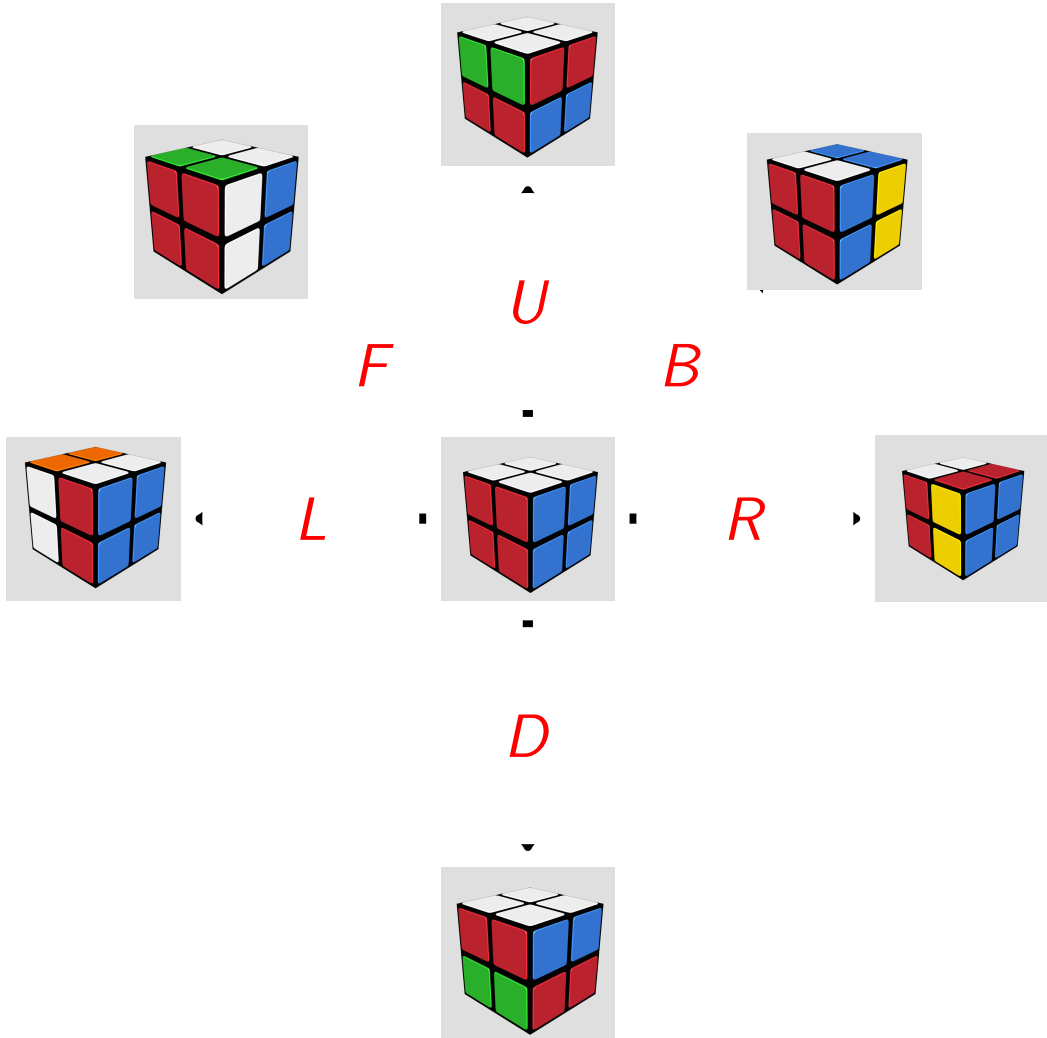
Cayley Graph: Local Neighborhood



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Cayley Graph: Local Neighborhood



Group Notation

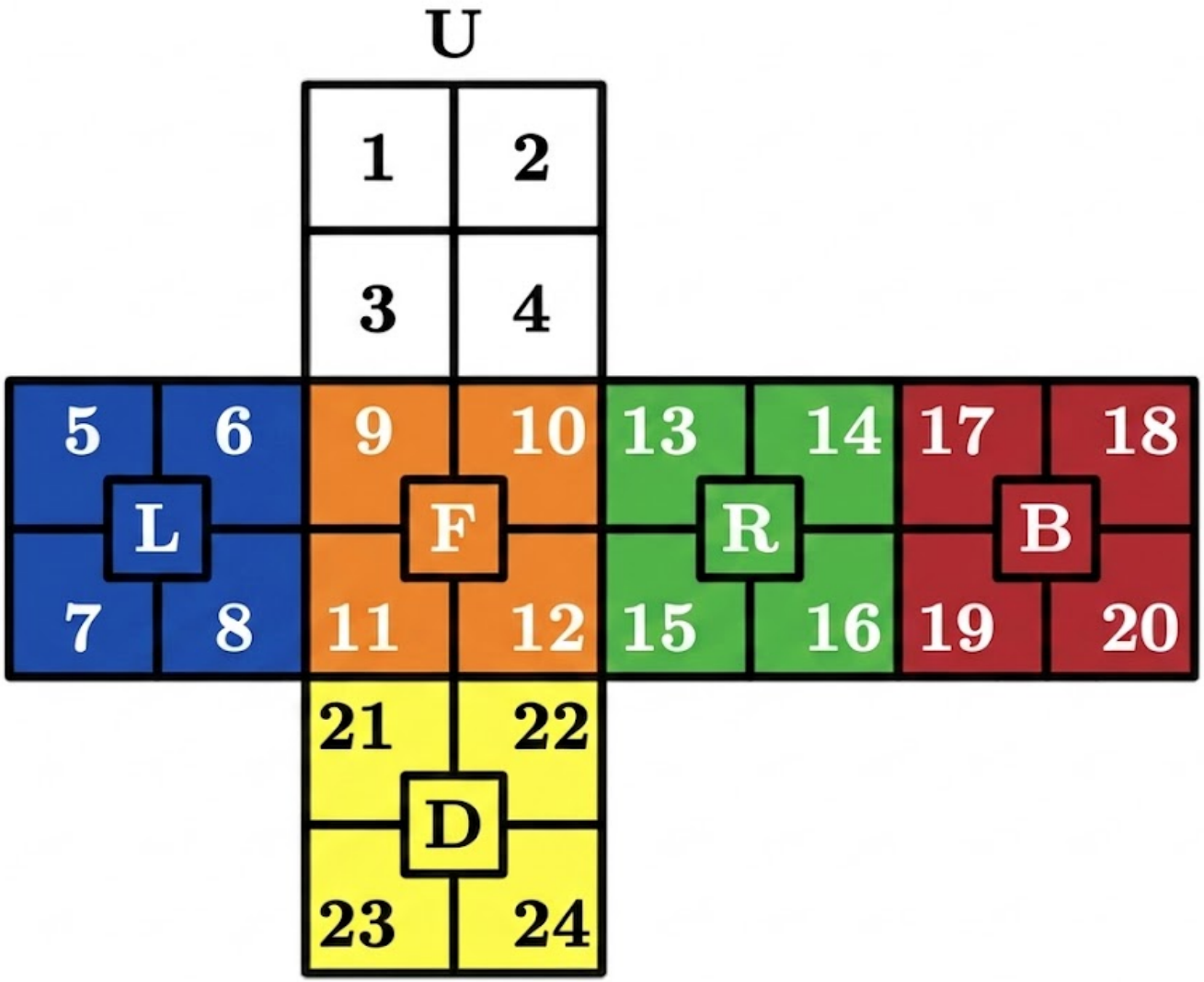
We first define the Rubik's Cube as a group. Let S_n denote the group of permutations of n elements under the composition operation $\circ$. For example, consider the permutation $\sigma = (2, 3, 1) \in S_3$. The permutation is defined by the mapping:

$$1 \mapsto 2, \quad 2 \mapsto 3, \quad 3 \mapsto 1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

The composition of two permutations, $\sigma \circ \tau$, is evaluated from right to left. For example:

$$(2, 3, 1) \circ (1, 3, 2) = (2, 1, 3)$$

Group Notation



Group Notation

The Cube has 24 stickers (4 per face on 6 faces). We define each state of the Cube with a permutation of its stickers.

- **Right Turn (R):**

$$R = \begin{pmatrix} 2 & 4 & 10 & 12 & 13 & 14 & 15 & 16 & 17 & 19 & 22 & 24 \\ 19 & 17 & 2 & 4 & 14 & 16 & 13 & 15 & 24 & 22 & 12 & 10 \end{pmatrix}$$

- **Up Turn (U):**

$$U = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 9 & 10 & 13 & 14 & 17 & 18 \\ 2 & 4 & 1 & 3 & 17 & 18 & 5 & 6 & 9 & 10 & 13 & 14 \end{pmatrix}$$

- **Front Turn (F):**

$$F = \begin{pmatrix} 3 & 4 & 6 & 8 & 9 & 10 & 11 & 12 & 13 & 15 & 21 & 22 \\ 13 & 15 & 4 & 3 & 10 & 12 & 9 & 11 & 22 & 21 & 6 & 8 \end{pmatrix}$$

Cayley Graphs

Let $\mathcal{R}$, the Rubik's Cube group, be the subgroup of S_{24} generated by the moves $\mathcal{R} =< \mathbf{R}, \mathbf{R}', \mathbf{U}, \mathbf{U}', \mathbf{F}, \mathbf{F}' >$.

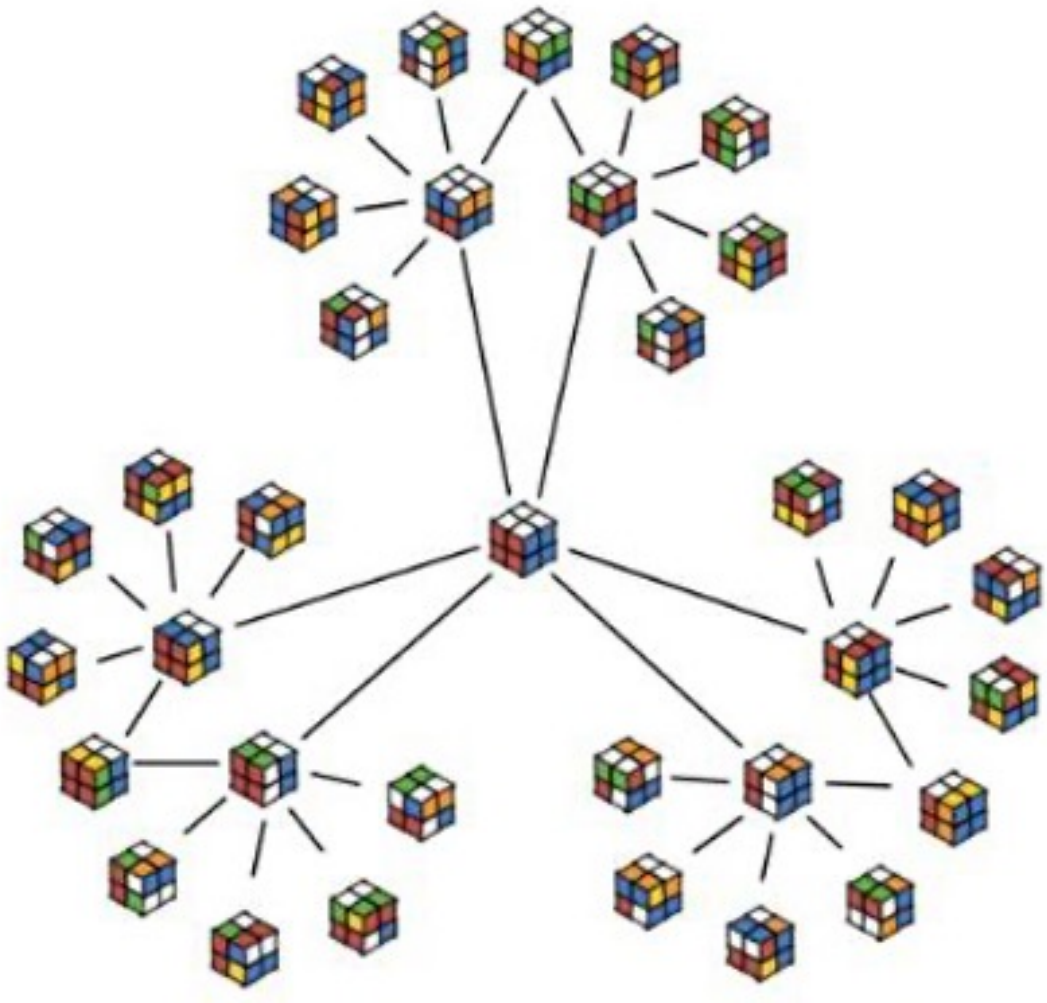
Definition 1 (Cayley Graph). Given a group $G = (A, *)$ and a set of generators $S \subseteq A$, the Cayley graph G_S is the graph whose vertex set is A and where two vertices $u, v \in G$ are adjacent if there exists a generator $s \in S$ such that $u * s = v$.

Let $G_{\mathcal{R}}$ denote the Cayley graph of the Rubik's cube group under the following set of generators:

$$S_{\mathcal{R}} := \{R, R', U, U', F, F'\}$$

Note: We use **Quarter-Turn Metric (QTM)**, which means 180 degree turns are counted as 2 moves.

Cayley Graphs



Hamiltonian Paths

Definition 2 (Hamiltonian Path). A Hamiltonian path in G is a path that contains every vertex of G.

For $G_{\mathcal{R}}$, we look for a Hamiltonian path from a scrambled state to the solved state. In practice, this means we move to every possible state of the Cube once before solving it.

Bipartite Structure

In the **Quarter-Turn Metric (QTM)**, the Cayley graph $G_{\mathcal{R}}$ is bipartite. Every generator is an odd permutation, meaning we alternate between two partitions:

- Partition A (Even): States reachable via an even number of turns.
  - Note: The *Solved State* $\in A$.
- Partition B (Odd): States reachable via an odd number of turns.

Hamiltonian Laceability

Definition 3 (Hamiltonian Laceability).
A bipartite graph is *Hamiltonian laceable* if for every $u \in A$ and $v \in B$, there exists a Hamiltonian path beginning at u and ending at v.

If $G_{\mathcal{R}}$ is laceable, then for any scramble $s \in B$, there exists a path of length $3,674,159$ that visits every state exactly once before solving the cube.

How do we prove Hamiltonian Laceability?

Option 1: Use a brute force algorithm to find path of unique 3+ million states. Computationally impossible.

Option 2: Utilize some of the facts of **Bipartite Cayley Graphs** to simplify this problem. In particular, Graph Automorphism and Vertex Transitivity.

Graph Automorphism

Definition 4. An automorphism over a graph $G = (V, E)$ is a bijection $\varphi : V \to V$ such that for any two vertices $u, v \in V$, we have that $(u, v) \in E$ if and only if $(\varphi(u), \varphi(v)) \in E$.

Given a state s of a Rubik's Cube, let $d(s)$ be the distance from s to the solved state.

Lemma. If φ is an automorphism over a graph $G = (V, E)$, then for any two vertices $u, v \in V$, we have $d(u, v) = d(\varphi(u), \varphi(v))$.

Vertex Transitivity

Definition 5 (Vertex Transitivity). A graph $G = (V, E)$ is vertex-transitive if for any two vertices $u, v \in V$, there exists an automorphism φ such that $\varphi(u) = v$.

Cayley Graphs and Vertex Transitivity

Lemma 3. Every Cayley graph is vertex-transitive.

Proof. Shown on the board.

Generalization of Laceability

Vertex transitivity allows us to fix one end of the path at the identity (solved state).

- Let e be the solved state.
- To prove $G_{\mathcal{R}}$ is laceable, it is sufficient to show that for every odd vertex $v \in B$, there exists a Hamiltonian path from v to e.
- Any pair $(u \in A, w \in B)$ can be mapped to a pair (e, w') via an automorphism φ that preserves the bipartite partitions.

References

- **Subercaseaux, B., et al.** (2025). *A Demigod's Number for the Rubik's Cube*. arXiv:2501.00144 [math.CO]. `https://arxiv.org/abs/2501.00144`
- **Quan, J.** (2025). *CMU Math Student Charts a New Path with Rubik's Cubes*. Carnegie Mellon University Department of Mathematical Sciences. `cmu.edu/math/news-events/articles/2025/1031`
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